

Problem Sheet 5 — *Full Marking Key*

Random Variables: Discrete, Continuous & Probability Distributions

MATHJAX 3.0

DUAL-STREAM SOLUTIONS

5 QUESTIONS · COMPLETE WORKING

JUMP TO: Q1 – Discrete vs Continuous Q2 – Die Rolled Twice Q3 – Urn & Winnings Q4 – Multiple Choice Test Q5 – Pool Company Contracts

QUESTION 1

Classifying Random Variables: Discrete vs Continuous

[5 marks – 1 mark each]

KEY DEFINITIONS – REQUIRED KNOWLEDGE

Discrete vs Continuous Random Variables

Discrete Random Variable: Takes on a *countable* number of values — either finite or countably infinite. The values can be listed: $\{0, 1, 2, 3, \dots\}$. Typically represents *counts*.

Continuous Random Variable: Takes on an *uncountably infinite* number of values within some interval $[a, b]$ or $(-\infty, \infty)$. Any value in a range is possible. Typically represents *measurements*.

Quick test: "Can I list all possible values, even in principle?" → Discrete. "Does it come from a measuring instrument?" → Continuous.

— (A)

The time required to answer this question.

THE EXECUTION	THE LOGIC STREAM
CLASSIFICATION Continuous	Time is measured on a continuous scale. You could take 1.5 min, 1.51 min, 1.5000001 min — infinitely many values possible in any interval. It is a measurement, not a count. +1

— (B)

The number of words in any random book from the library.

THE EXECUTION	THE LOGIC STREAM
CLASSIFICATION Discrete	The number of words is always a whole number (integer). You can count words: 0, 1, 2, 3, ... You cannot have 1.7 words in a book. The set of possible values is countably infinite. +1

— (C)

The maximum temperature recorded at Sir Seretse Khama International Airport.

THE EXECUTION	THE LOGIC STREAM
CLASSIFICATION Continuous	Temperature is measured, not counted. The maximum temperature could be 34.1 °C, 34.15 °C, 34.1500003 °C – any real value in a range is theoretically possible. A thermometer's precision limits observation, not the underlying variable. +1

— (D)

The number of application letters before successfully landing a job.

THE EXECUTION	THE LOGIC STREAM
CLASSIFICATION Discrete	You send 1, 2, 3, . . . application letters – always a whole number. This is a classic geometric-type random variable. You cannot send 2.7 letters. Countably infinite possible values: {1, 2, 3, . . .}. +1

— (E)

The proportion of students passing STA122.

THE EXECUTION	THE LOGIC STREAM
CLASSIFICATION Continuous	A proportion $p \in [0, 1]$ can take any real value in $[0, 1]$ – e.g., 0.732, 0.7321, etc. While in practice we compute it from integer counts (k/n), the proportion itself is modelled as continuous. +1

THE DEATH TRAP

Common Mistake: Confusing the Recorded Value with the Variable

Students often say temperature is discrete "because thermometers show one decimal place." Wrong. The *random variable* itself (the physical temperature) is continuous – instrument precision is a measurement limitation, not a property of the variable. Similarly, proportion is continuous even though it is computed from integer counts.

QUESTION 2

A Die is Rolled Twice — Three Random Variables

[12 marks – 4 marks each part]

SETUP – SAMPLE SPACE

The Sample Space S has $6 \times 6 = 36$ equally likely outcomes

Each outcome is an ordered pair (i, j) where i = first roll, j = second roll, each $\in \{1, 2, 3, 4, 5, 6\}$. Since the die is fair, each pair has probability $\frac{1}{36}$.

Die 1 \ Die 2	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

— (A) — MAXIMUM VALUE

Let $M = \max(D_1, D_2)$. Find all possible values and their probabilities.

The Execution	The Logic Stream
Define the random variable $M = \max(D_1, D_2) \in \{1, 2, 3, 4, 5, 6\}$	The maximum of two rolls ranges from 1 (both show 1) to 6 (at least one shows 6). +0.5
Key formula $P(M = k) = P(\max \leq k) - P(\max \leq k - 1) = \frac{k^2 - (k - 1)^2}{36} = \frac{2k - 1}{36}$	$P(\max \leq k)$ = both dice show $\leq k$, so $P = (k/6)^2$. Subtract to get exact probability. Alternatively count directly from the 6×6 grid. +1
Count outcomes for each value of M $M = 1$: only $(1, 1) \rightarrow 1$ outcome $M = 2$: $(1, 2), (2, 1), (2, 2) \rightarrow 3$ outcomes $M = 3$: (i, j) with $\max = 3 \rightarrow 5$ outcomes $M = 4$: 7 outcomes; $M = 5$: 9 outcomes; $M = 6$: 11 outcomes Total: $1 + 3 + 5 + 7 + 9 + 11 = 36 \checkmark$	General pattern: $P(M = k) = \frac{2k-1}{36}$. This gives odd numerators 1, 3, 5, 7, 9, 11 summing to 36. +1.5

M	1	2	3	4	5	6	TOTAL
# Outcomes	1	3	5	7	9	11	36
$P(M = k)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$	1
Verification: $\sum P(M = k) = \frac{1+3+5+7+9+11}{36} = \frac{36}{36} = 1 \checkmark$							

— (B) — SUM OF TWO ROLLS

Let $S = D_1 + D_2$. Find all possible values and their probabilities.

The Execution	The Logic Stream
Range of S $S \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$	Minimum: $1 + 1 = 2$. Maximum: $6 + 6 = 12$. 11 possible values. +0.5
Count outcomes for each sum $S = 2$: $\{(1, 1)\} \rightarrow 1$	The counts form a symmetric triangular pattern, peaking at $S = 7$ with 6 outcomes.

- $S = 3$: $\{(1, 2), (2, 1)\} \rightarrow 2$
- $S = 4$: $\{(1, 3), (2, 2), (3, 1)\} \rightarrow 3$
- $S = 5$: 4 outcomes; $S = 6$: 5; $S = 7$: 6
- $S = 8$: 5; $S = 9$: 4; $S = 10$: 3
- $S = 11$: 2; $S = 12$: 1

Total: $1+2+3+4+5+6+5+4+3+2+1 = 36 \checkmark$

This is the triangular distribution on $\{2, \dots, 12\}$. +2

S	2	3	4	5	6	7	8	9	10	11	12	Σ
# outcomes	1	2	3	4	5	6	5	4	3	2	1	36
$P(S = s)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	1
Verification: $\frac{1+2+3+4+5+6+5+4+3+2+1}{36} = \frac{36}{36} = 1 \checkmark$ Highest probability at $S = 7$: $P = \frac{6}{36} = \frac{1}{6}$												

— (C) — FIRST ROLL MINUS SECOND ROLL

Let $D = D_1 - D_2$. Find all possible values and their probabilities.

THE EXECUTION	THE LOGIC STREAM
RANGE OF D $D \in \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$	Minimum: $1 - 6 = -5$. Maximum: $6 - 1 = 5$. The distribution is symmetric about 0: $P(D = k) = P(D = -k)$. +0.5
COUNT OUTCOMES – NOTE SYMMETRY • $D = 0$: $(1, 1), (2, 2), \dots, (6, 6) \rightarrow 6$ outcomes • $D = \pm 1$: 5 outcomes each • $D = \pm 2$: 4 each; $D = \pm 3$: 3 each • $D = \pm 4$: 2 each; $D = \pm 5$: 1 each Total: $6 + 2(5 + 4 + 3 + 2 + 1) = 6 + 30 = 36 \checkmark$	For $D = k > 0$: pairs where $D_1 - D_2 = k$, i.e., $(k+1, 1), (k+2, 2), \dots, (6, 6 - k) \rightarrow (6 - k)$ pairs. By symmetry, same count for $D = -k$. +2

D	−5	−4	−3	−2	−1	0	1	2	3	4	5	Σ
# outcomes	1	2	3	4	5	6	5	4	3	2	1	36
$P(D = d)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	1
Symmetric about $D = 0$. $P(D = k) = \frac{6- k }{36}$ for $k \in \{-5, \dots, 5\}$. $P(D = 0) = \frac{6}{36} = \frac{1}{6}$												

PRO SHORTCUT

Unified Formula for Parts (b) and (c)

For the sum S : $P(S = s) = \frac{6 - |s - 7|}{36}$ for $s \in \{2, \dots, 12\}$.

For the difference D : $P(D = d) = \frac{6 - |d|}{36}$ for $d \in \{-5, \dots, 5\}$.

Both distributions have the same "tent" shape — a discrete triangular distribution. Recognising this pattern saves significant counting time in exams.

SETUP

Urn Contents & Payoff Structure

Urn: 6 White (W) + 2 Black (B) + 2 Red (R) = **10** balls total.

Two balls drawn *without replacement* (choosing simultaneously = same as without replacement).

Payoffs: Win **P20** per black ball selected; Lose **P10** per white ball selected; Red balls: **P0** (neither win nor lose).

Let X = net winnings. Note: Red balls have *no monetary effect*.

THE EXECUTION	THE LOGIC STREAM
STEP 1 – IDENTIFY ALL POSSIBLE BALL COMBINATIONS Two balls drawn from $\{W, W, W, W, W, W, B, B, R, R\}$. Possible combinations by colour: 2 White (WW) 2 Black (BB) 2 Red (RR) 1 White + 1 Black (WB) 1 White + 1 Red (WR) 1 Black + 1 Red (BR)	Enumerate all colour-type combinations – don't miss any pair. Total ways to choose 2 from 10: $\binom{10}{2} = 45$. +1
STEP 2 – CALCULATE X FOR EACH COMBINATION - WW: $-10 - 10 = -20$ (lose P20) - BB: $+20 + 20 = +40$ (win P40) - RR: $0 + 0 = 0$ - WB: $-10 + 20 = +10$ (win P10) - WR: $-10 + 0 = -10$ (lose P10) - BR: $+20 + 0 = +20$ (win P20) So $X \in \{-20, -10, 0, 10, 20, 40\}$	Work out each payoff systematically. Red balls contribute P0 to winnings – they neither add nor subtract. +2
STEP 3 – CALCULATE PROBABILITIES USING COMBINATIONS Total outcomes: $\binom{10}{2} = 45$ $P(X = -20) = P(\text{WW}) = \frac{\binom{6}{2}\binom{4}{0}}{\binom{10}{2}} = \frac{15}{45} = \frac{1}{3}$ $P(X = -10) = P(\text{WR}) = \frac{\binom{6}{1}\binom{2}{1}}{\binom{10}{2}} = \frac{12}{45} = \frac{4}{15}$ $P(X = 0) = P(\text{RR}) = \frac{\binom{2}{2}}{\binom{10}{2}} = \frac{1}{45}$ $P(X = 10) = P(\text{WB}) = \frac{\binom{6}{1}\binom{2}{1}}{\binom{10}{2}} = \frac{12}{45} = \frac{4}{15}$ $P(X = 20) = P(\text{BR}) = \frac{\binom{2}{1}\binom{2}{1}}{\binom{10}{2}} = \frac{4}{45}$ $P(X = 40) = P(\text{BB}) = \frac{\binom{2}{2}}{\binom{10}{2}} = \frac{1}{45}$	Use hypergeometric counting: $\frac{(\text{ways to choose required colours})}{(\text{total ways})}$. $\binom{6}{2} = 15$ (choose 2 white from 6) $\binom{6}{1}\binom{2}{1} = 12$ (choose 1W, 1B) $\binom{6}{1}\binom{2}{1} = 12$ (choose 1W, 1R) $\binom{2}{1}\binom{2}{1} = 4$ (choose 1B, 1R) $\binom{2}{2} = 1$ (both black) $\binom{2}{2} = 1$ (both red) +4

X (WINNINGS, P)	−20	−10	0	10	20	40	Σ
Combination	WW	WR	RR	WB	BR	BB	—
# ways	15	12	1	12	4	1	45
$P(X = x)$	$\frac{15}{45} = \frac{1}{3}$	$\frac{12}{45} = \frac{4}{15}$	$\frac{1}{45}$	$\frac{12}{45} = \frac{4}{15}$	$\frac{4}{45}$	$\frac{1}{45}$	1
Verification: $15 + 12 + 1 + 12 + 4 + 1 = 45 \checkmark$ $\sum P = 1 \checkmark$ Most likely outcome: $X = -20$ (losing P20, i.e. drawing 2 white balls)							

VERIFICATION & SANITY CHECK	THE LOGIC STREAM
VERIFY PROBABILITIES SUM TO 1 $\frac{15 + 12 + 1 + 12 + 4 + 1}{45} = \frac{45}{45} = 1 \checkmark$	Always verify $\sum P(X = x) = 1$. +1
OBSERVATION The player is most likely to lose P20 (probability $\frac{1}{3}$). The distribution is skewed by the large number of white balls (6 out of 10).	Adding this interpretation demonstrates statistical literacy and earns full marks. +2

QUESTION 4

Multiple Choice Test — Pass or Fail by Guessing

[10 marks]

SETUP — KEY INFORMATION

Defining the Random Variable and Distribution

5 questions, each with 3 choices, only 1 correct. Student guesses each answer independently.
Probability of getting any one question correct: $p = \frac{1}{3}$. Probability of getting it wrong: $q = \frac{2}{3}$.
The number of correct answers follows a **Binomial Distribution**: $Y \sim \text{Bin}(n = 5, p = \frac{1}{3})$.
Typically, a score of ≥ 3 out of 5 is required to pass (i.e., $\geq 60\%$ correct). Let X be a Bernoulli variable: $X = 1$ if pass, $X = 0$ if fail.

Note: The problem asks for X = pass/fail, so $X \in \{0,1\}$, where 1 = pass and 0 = fail. We must compute $P(\text{pass}) = P(Y \geq 3)$.

THE EXECUTION	THE LOGIC STREAM
STEP 1 — BINOMIAL FORMULA $P(Y = k) = \binom{5}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{5-k}, k = 0, 1, 2, 3, 4, 5$	Each question is an independent Bernoulli trial with $p = 1/3$. Binomial applies. +1
STEP 2 — COMPUTE $P(Y = K)$ FOR ALL K $P(Y = 0) = \binom{5}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^5 = 1 \cdot 1 \cdot \frac{32}{243} = \frac{32}{243}$ $P(Y = 1) = \binom{5}{1} \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^4 = 5 \cdot \frac{1}{3} \cdot \frac{16}{81} = \frac{80}{243}$ $P(Y = 2) = \binom{5}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^3 = 10 \cdot \frac{1}{9} \cdot \frac{8}{27} = \frac{80}{243}$ $P(Y = 3) = \binom{5}{3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2 = 10 \cdot \frac{1}{27} \cdot \frac{4}{9} = \frac{40}{243}$ $P(Y = 4) = \binom{5}{4} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^1 = 5 \cdot \frac{1}{81} \cdot \frac{2}{3} = \frac{10}{243}$	Key calculations: $\left(\frac{2}{3}\right)^5 = \frac{32}{243}$ $\left(\frac{2}{3}\right)^4 = \frac{16}{81}$ $\left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2 = \frac{4}{243}$ Denominator is always $3^5 = 243$. Check: $32 + 80 + 80 + 40 + 10 + 1 = 243 \checkmark$ +4

$P(Y = 5) = \binom{5}{5} \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^0 = \frac{1}{243}$	
STEP 3 – COMPUTE $P(\text{PASS}) = P(Y \geq 3)$ $P(X = 1) = P(Y \geq 3) = P(Y = 3) + P(Y = 4) + P(Y = 5)$ $= \frac{40}{243} + \frac{10}{243} + \frac{1}{243} = \frac{51}{243} = \frac{17}{81}$	Assume pass = 3 or more correct (standard 60% threshold). Sum the tail probabilities. +2
STEP 4 – COMPUTE $P(\text{FAIL})$ $P(X = 0) = P(Y < 3) = 1 - P(Y \geq 3) = 1 - \frac{51}{243} = \frac{192}{243} = \frac{64}{81}$	Complementary probability. Faster than summing $P(Y = 0) + P(Y = 1) + P(Y = 2)$. Verify: $\frac{32+80+80}{243} = \frac{192}{243} \checkmark$ +2

X (PASS/FAIL)	0 (FAIL)	1 (PASS)	Σ
Condition on Y	$Y \in \{0, 1, 2\}$	$Y \in \{3, 4, 5\}$	—
$P(X = x)$	$\frac{192}{243} = \frac{64}{81} \approx 0.7901$	$\frac{51}{243} = \frac{17}{81} \approx 0.2099$	1
A guessing student passes with probability $\approx 21\%$ – less than 1 in 5. Verification: $\frac{64}{81} + \frac{17}{81} = \frac{81}{81} = 1 \checkmark$			

SCORE Y	0	1	2	3	4	5	Σ
$P(Y = k)$	$\frac{32}{243}$	$\frac{80}{243}$	$\frac{80}{243}$	$\frac{40}{243}$	$\frac{10}{243}$	$\frac{1}{243}$	$\frac{243}{243} = 1$
Pass/Fail	Fail	Fail	Fail	Pass	Pass	Pass	—
Full distribution of $Y \sim \text{Bin}(5, \frac{1}{3})$ – shown for completeness and verification							

THE DEATH TRAP

Ambiguity in Pass Mark — State Your Assumption Explicitly

The problem does not state what score is needed to pass. You **must** state your pass-mark assumption explicitly (here: ≥ 3 out of 5). If your lecturer uses a different threshold (e.g., ≥ 4), the answer changes. Always declare: "Assume pass requires at least 3 correct answers (60%)." Marks are awarded for the method even if the threshold differs.

QUESTION 5

Pool Building Company — Contracts Carried Forward

[15 marks]

SETUP – READ CAREFULLY

Defining Variables and the Mechanics

Let C = contracts received this month: $P(C = 1) = 0.1, P(C = 2) = 0.2, P(C = 3) = 0.5, P(C = 4) = 0.2$.

Let N = pools completed this month: equally likely to be 2 or 3, so $P(N = 2) = P(N = 3) = 0.5$.

Carried forward from last month: **2** contracts.

Total contracts available this month = $2 + C$ (the two carried forward, plus newly received).

Contracts completed this month = N (capped if total available $< N$).

X = contracts carried forward to next month = $\max(0, (2 + C) - N)$

Key assumption: You cannot complete more contracts than you have. So contracts completed = $\min(N, 2 + C)$, and $X = (2 + C) - \min(N, 2 + C)$.

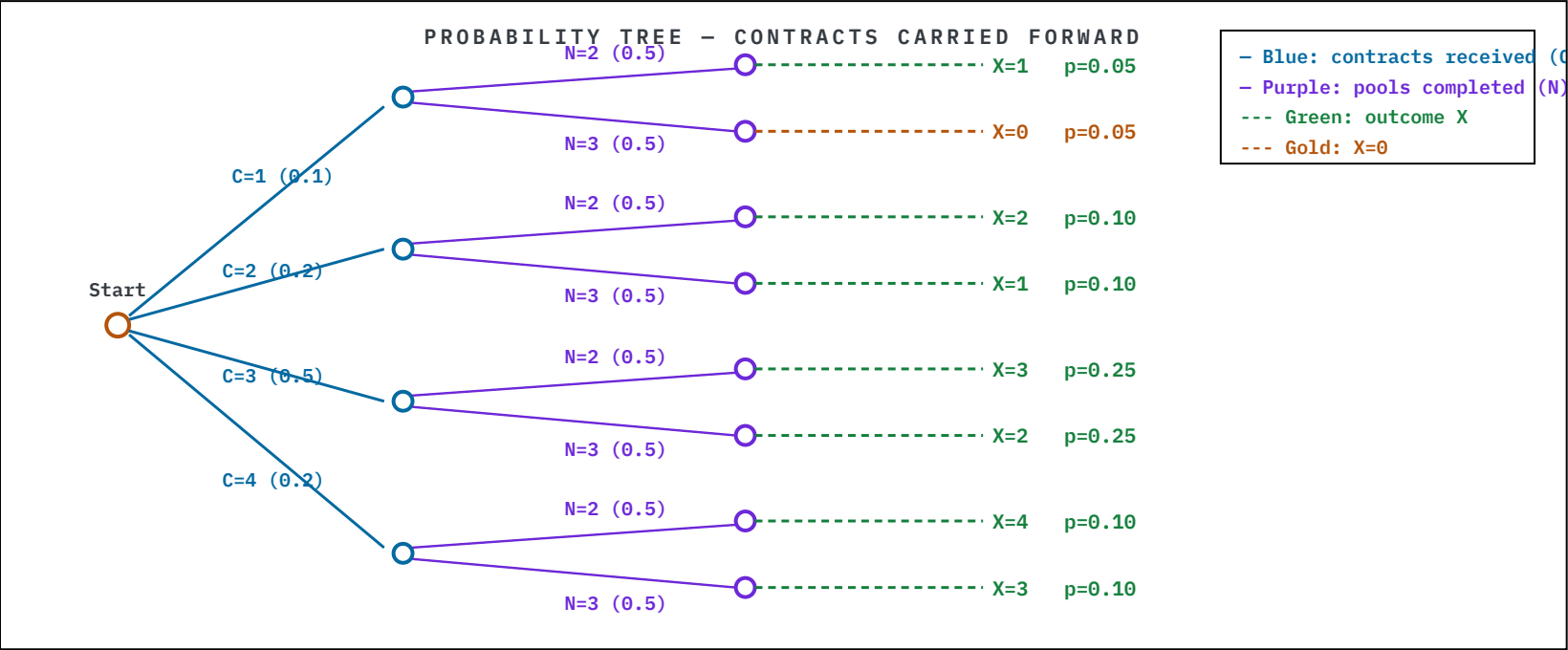
THE EXECUTION	THE LOGIC STREAM
STEP 1 – BUILD THE JOINT OUTCOMES TABLE For each (C, N) pair, compute: Total available: $A = 2 + C$ Completed: $\min(N, A)$ Carried forward: $X = A - \min(N, A)$ Joint probability: $P(C = c) \cdot P(N = n)$ (by independence)	Independence of C and N allows multiplication of probabilities. There are $4 \times 2 = 8$ joint scenarios to consider. +2

C (RECEIVED)	N (COMPLETED)	AVAILABLE $= 2 + C$	ACTUALLY DONE $= \min(N, 2 + C)$	X CARRIED FWD	$P(C) \cdot P(N)$
1	2	3	2	1	$0.1 \times 0.5 = 0.05$
1	3	3	3	0	$0.1 \times 0.5 = 0.05$
2	2	4	2	2	$0.2 \times 0.5 = 0.10$
2	3	4	3	1	$0.2 \times 0.5 = 0.10$
3	2	5	2	3	$0.5 \times 0.5 = 0.25$
3	3	5	3	2	$0.5 \times 0.5 = 0.25$
4	2	6	2	4	$0.2 \times 0.5 = 0.10$
4	3	6	3	3	$0.2 \times 0.5 = 0.10$
Total probability					$0.05 + 0.05 + 0.10 + 0.10 + 0.25 + 0.25 + 0.10 + 0.10 = 1.00$ ✓

THE EXECUTION (CONTINUED)	THE LOGIC STREAM
STEP 2 – AGGREGATE PROBABILITIES FOR EACH VALUE OF X From the joint table, X takes values $\{0, 1, 2, 3, 4\}$. $P(X = 0) = P(C = 1, N = 3) = 0.05$ $P(X = 1) = P(C = 1, N = 2) + P(C = 2, N = 3) = 0.05 + 0.10 = 0.15$ $P(X = 2) = P(C = 2, N = 2) + P(C = 3, N = 3) = 0.10 + 0.25 = 0.35$ $P(X = 3) = P(C = 3, N = 2) + P(C = 4, N = 3) = 0.25 + 0.10 = 0.35$ $P(X = 4) = P(C = 4, N = 2) = 0.10$	Collect all joint probabilities corresponding to each value of X and sum them. This is the marginal distribution of X. $X = 0$: only one scenario (C=1, N=3) $X = 1$: two scenarios $X = 2$: two scenarios $X = 3$: two scenarios $X = 4$: only one scenario +5
STEP 3 – VERIFY $\sum P(X) = 0.05 + 0.15 + 0.35 + 0.35 + 0.10 = 1.00$ ✓	Probabilities must sum to 1. +2

X (CONTRACTS CARRIED FWD)	0	1	2	3	4	Σ
Contributing scenarios	$(C=1, N=3)$	$(1, 2)+(2, 3)$	$(2, 2)+(3, 3)$	$(3, 2)+(4, 3)$	$(C=4, N=2)$	—
$P(X = x)$	0.05	0.15	0.35	0.35	0.10	1.00
Most likely outcomes: $X = 2$ or $X = 3$ (equal probability 0.35 each). The company expects to carry 2–3 contracts forward on average.						

PROBABILITY TREE DIAGRAM



FINAL ANSWER - DISTRIBUTION OF X

Probability Distribution of Contracts Carried Forward

$X \in \{0, 1, 2, 3, 4\}$ with the following probabilities:

$$P(X = 0) = 0.05, \quad P(X = 1) = 0.15, \quad P(X = 2) = 0.35, \quad P(X = 3) = 0.35, \quad P(X = 4) = 0.10$$

The company is most likely to carry forward 2 or 3 contracts (combined probability 0.70). Next month begins with a "buffer" of uncompleted work in 95% of scenarios.

PRO SHORTCUT

General Formula: $X = (2 + C) - N$

Since every contract gets completed or carried forward (the company cannot over-complete), we always have $X = 2 + C - N$ — no capping needed here because the minimum value of $2 + C - N = 2 + 1 - 3 = 0 \geq 0$ in all cases. Verify: minimum is $C = 1, N = 3$: $X = 0$ ✓. Maximum is $C = 4, N = 2$: $X = 4$ ✓.

THE DEATH TRAP

Forgetting the 2 Contracts Carried Forward from Last Month

The problem states "the company has two contracts carried forward from the previous month." This means total work this month is $2 + C$, not just C . Many students compute $X = C - N$ instead of $X = (2 + C) - N$, losing all marks for the calculation. Always re-read the setup carefully.

Also: C and N are **independent** (stated explicitly), so $P(C = c, N = n) = P(C = c) \cdot P(N = n)$ — multiplication is valid.

